

NORMANDIN, QC, Canada

WMO: 713790

Lat: 48.842N

Lon: 72.547W

Elev: 137

StdP: 99.69

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-33.5	-30.5	-37.6	0.1	-33.3	-34.6	0.1	-30.3	14.0	-14.7	12.7	-14.6	2.6	310	0.681

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.6	28.5	19.8	26.6	19.2	24.9	18.4	21.6	26.2	20.5	24.7	19.5	23.3	5.1	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.1	15.0	23.9	19.0	14.0	22.9	18.0	13.1	21.6	63.9	26.3	60.0	25.0	56.3	23.4	27.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
11.4	10.0	8.8	-37.8	31.7	3.0	2.2	-40.0	33.3	-41.7	34.6	-43.4	35.8	-45.6	37.4		
			-38.0	23.7	2.9	1.2	-40.1	24.6	-41.8	25.3	-43.5	25.9	-45.6	26.8		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	2.0	-16.8	-15.1	-8.2	0.6	9.5	15.0	17.7	16.4	11.9	5.1	-2.4	-11.3	(6)
(7)		DBStd	13.23	7.97	7.67	7.05	4.94	4.39	4.03	2.97	3.13	4.26	4.17	5.66	7.40	(7)
(8)		HDD10.0	3667	829	702	564	281	63	5	0	1	27	161	372	661	(8)
(9)		HDD18.3	6039	1088	936	822	531	275	114	47	77	198	410	622	919	(9)
(10)		CDD10.0	737	0	0	0	1	49	156	240	198	83	9	1	0	(10)
(11)		CDD18.3	66	0	0	0	0	2	15	28	17	4	0	0	0	(11)
(12)		CDH23.3	734	0	0	0	0	52	201	273	162	44	2	0	0	(12)
(13)		CDH26.7	159	0	0	0	0	12	52	60	29	7	0	0	0	(13)
(14)	Wind	WSAvg	3.8	4.1	4.1	4.4	4.3	4.0	3.6	3.3	3.2	3.4	3.8	4.0	3.9	(14)
(15)	Precipitation	PrecAvg	913	62	48	48	52	81	81	113	101	94	67	66	78	(15)
(16)		PrecMax	1132	146	95	102	140	223	158	228	197	154	155	125	147	(16)
(17)		PrecMin	663	22	8	5	3	9	32	56	42	42	17	13	33	(17)
(18)		PrecStd	114	28	28	25	27	45	34	41	40	31	29	32	34	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	4.1	4.1	7.3	17.3	28.3	31.0	30.6	29.6	27.4	20.9	13.0	3.8	(19)
(20)			MCWB	3.0	2.8	3.9	9.8	16.4	20.4	21.1	21.0	20.4	15.5	10.7	2.6	(20)
(21)		2%	DB	1.5	1.5	4.8	13.6	24.4	27.9	28.3	27.1	24.3	17.2	9.2	2.1	(21)
(22)			MCWB	1.0	0.3	2.1	7.5	15.1	18.9	20.3	20.1	19.0	13.4	7.2	1.3	(22)
(23)		5%	DB	-1.1	-0.2	3.1	10.3	21.1	25.7	26.3	25.1	21.7	14.3	6.7	0.7	(23)
(24)			MCWB	-1.7	-1.0	1.2	5.7	13.5	17.9	19.5	19.0	17.2	11.4	5.3	0.1	(24)
(25)		10%	DB	-4.3	-2.6	1.8	7.6	18.3	23.2	24.4	23.2	19.5	11.8	4.6	-0.5	(25)
(26)			MCWB	-4.9	-3.5	0.4	4.1	12.1	16.6	18.6	17.9	15.9	9.5	3.5	-1.1	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.3	3.2	4.9	10.9	18.9	22.4	23.1	22.4	21.6	16.5	11.3	3.0	(27)
(28)			MCDB	3.8	4.1	6.6	15.8	25.1	28.2	28.0	27.2	26.2	20.1	12.7	3.7	(28)
(29)		2%	WB	1.1	0.6	2.6	8.1	16.6	20.7	21.7	21.1	19.8	13.9	7.7	1.4	(29)
(30)			MCDB	1.4	1.3	4.1	12.6	21.0	25.4	25.8	25.5	23.3	16.9	8.9	2.0	(30)
(31)		5%	WB	-1.8	-1.1	1.5	6.1	14.4	19.3	20.6	19.9	18.1	11.7	5.4	0.2	(31)
(32)			MCDB	-1.2	-0.2	2.8	9.6	19.8	23.6	24.7	23.8	21.0	13.6	6.5	0.6	(32)
(33)		10%	WB	-5.0	-3.4	0.5	4.4	12.8	17.8	19.4	18.7	16.4	9.9	3.6	-1.1	(33)
(34)			MCDB	-4.4	-2.6	1.8	7.1	17.7	22.0	23.2	22.0	19.1	11.7	4.6	-0.5	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	11.4	12.5	12.0	10.1	12.7	12.5	11.6	11.7	11.3	8.7	7.5	9.3	(35)
(36)			MCDBR	11.0	10.7	10.8	14.1	18.0	16.2	14.3	14.6	14.4	13.2	9.6	7.7	(36)
(37)			MCWBR	10.7	9.8	8.9	9.0	9.9	8.5	7.8	8.3	9.1	9.5	8.0	7.5	(37)
(38)		5% WB	MCDBR	10.8	10.0	9.5	13.1	15.1	13.6	12.6	12.3	12.9	11.6	8.6	7.8	(38)
(39)			MCWBR	10.5	9.4	8.1	8.9	9.4	8.1	7.5	7.6	8.8	8.8	7.6	7.6	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.271	0.285	0.303	0.376	0.355	0.384	0.400	0.377	0.366	0.331	0.322	0.264	(40)	
(41)		taud	2.229	2.213	2.231	2.227	2.423	2.366	2.342	2.406	2.424	2.498	2.344	2.309	(41)	
(42)		Ebn at Noon	783	863	909	869	905	877	856	861	834	811	715	750	(42)	
(43)		Edh at Noon	75	97	114	128	110	118	119	107	95	75	69	61	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.32	2.34	3.66	4.63	5.23	5.61	5.14	4.36	3.32	1.91	1.02	0.84	(44)	
(45)		RadStd	0.17	0.30	0.34	0.39	0.48	0.41	0.50	0.28	0.28	0.19	0.12	0.10	(45)	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(47)	Regional (16 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page